

SHAFAYET KHAN SHAFEE

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☎ +880-1791-051104 📍 Dhaka, Bangladesh

Statistician and data scientist applying rigorous statistical methods to public health research and industry problems, with a commitment to reproducible research and open-source software development.

RESEARCH INTERESTS

Causal Inference, Medical Statistics, Epidemiology, Health Data Science, Causal Mediation Analysis, Survival Analysis, Hierarchical Modeling, Bayesian Inference, Statistical Machine Learning.

EDUCATION

M.Sc. in Applied Statistics

GPA 3.97 / 4.00

Institute of Statistical Research & Training

University of Dhaka

Dhaka, Bangladesh

2022–2023

- **Thesis:** Developed point and interval estimators extending the median odds ratio framework for quantifying between-cluster variation in three-level hierarchical binary data, evaluated their statistical performance through extensive Monte Carlo simulation studies, and implemented the methodology in an open-source R package for post-estimation inference.
- **Selected Coursework:** Causal Inference; Multilevel Modeling; Bayesian Inference; Spatial Statistics; Statistical Machine Learning.

B.Sc. in Applied Statistics

CGPA 3.96 / 4.00

Institute of Statistical Research & Training

University of Dhaka

Dhaka, Bangladesh

2018–2022

- **Project:** Examined the association between caregivers' stimulation activities and socio-emotional development in under-five children using nationally representative MICS 2019 Bangladesh data and survey-weighted logistic regression analysis.
- **Selected Coursework:** Statistical Inference; Sampling Methods; Multivariate Statistics; Generalized Linear Models; Analysis of Time Series; Design & Analysis of Experiments; Epidemiology; Survival Analysis; Econometrics; Actuarial Statistics; Industrial Statistics & Operations Research.

PROFESSIONAL EXPERIENCE

Data Scientist

Pathao Limited

Dhaka, Bangladesh

Jul 2023–Present

- Applied Bayesian experimental design (Bayes Factor Design Analysis) to estimate sample sizes for controlled product experiments, incorporating prior information and evidence-based stopping criteria.
- Designed and analyzed randomized controlled experiments, including randomization checks through covariate balance assessment and Sample Ratio Mismatch (SRM) testing, to evaluate business initiatives.
- Estimated causal effects of a fintech product intervention on merchant transaction behavior using propensity score matching and Difference-in-Differences.

- Applied Hierarchical Bayesian modeling to evaluate adoption across cohorts in a controlled experiment, accounting for small and unequal group sizes and quantifying uncertainty through posterior distributions.
- Applied causal ML methods, namely Double ML and generalized random forests, to estimate heterogeneous treatment effects across user subgroups, supporting targeted intervention design.
- Co-developed a repayment risk scoring framework using Random Survival Forests to model time-to-repayment distributions, drawing on survival analysis methods for a credit risk application.
- Built scalable analytical pipelines using dbt and BigQuery, supporting reproducible statistical workflows and standardized automated analytical reporting.

PUBLICATIONS

† Corresponding author * Equal contribution

- [1] **S. K. Shafee**[†] and M. S. Rahman. *On the estimation of the median odds ratio for measuring contextual effects in multilevel binary data from complex survey designs*. [Under review]. June 2026. arXiv: 2606.15145 [stat.ME]. URL: <https://arxiv.org/abs/2606.15145>.
- [2] **S. K. Shafee**[†], B. Sarker^{*}, and M. N. I. Sium^{*}. *G-computation for causal effect estimation from observational hierarchical data with unmeasured cluster context*. [Under review]. June 2026. arXiv: 2606.14131 [stat.ME]. URL: <https://arxiv.org/abs/2606.14131>.
- [3] M. N. I. Sium, **S. K. Shafee**[†], M. H. Patwary, and B. Sarker. *Quantifying Inconsistent Mediation: The Mediated Magnitude Share and the Degree of Suppression*. [Under review]. Sept. 2026. arXiv: 2609.33846 [stat.ME]. URL: <https://arxiv.org/abs/2609.33846>.
- [4] **S. K. Shafee**[†], M. N. I. Sium, B. Sarker, and R. Islam. “Investigating the causal effect of maternal continuum of care on child’s minimum acceptable diet: A multilevel approach using partially pooled propensity score weighting”. In: *PLOS ONE* 20.11 (Nov. 2025), pp. 1–13. DOI: [10.1371/journal.pone.0335972](https://doi.org/10.1371/journal.pone.0335972).

AWARDS & ACHIEVEMENTS

- **Dean’s Award**, *Faculty of Science, University of Dhaka* **2025**
- **Conference Award for Scientists**, for the abstract “Interval Estimation of the Median Odds Ratio for Measuring Contextual Effects in Multilevel Data Using a Binary Logistic Model”, *45th Annual Conference of the International Society for Clinical Biostatistics (ISCB45), Thessaloniki, Greece* **2024**
- **National Science and Technology (NST) Fellowship**, for M.Sc. thesis research on multilevel modeling, *Ministry of Science and Technology (MoST), Government of Bangladesh* **2023**

TALKS & PRESENTATIONS

- “On the Estimation of the Median Odds Ratio for Measuring Contextual Effects in Multilevel Binary Data from Complex Survey Designs”, *International Conference on Applied Statistics and Data Science (ICASDS 2025), University of Dhaka, Bangladesh* **2025**

OPEN SOURCE CONTRIBUTIONS

R PACKAGES

- **MOR** — Post-estimation functions to compute the Median Odds Ratio and corresponding confidence interval from fitted multilevel binary logistic regression models.
- **skmisc** — Miscellaneous utility R functions.

PYTHON PACKAGES

- [skmiscpy](#) — Reusable functions for causal inference diagnostics, including mirror histograms, Love plots, and standardized mean difference (SMD) computation for inverse probability weighting (IPW).
- [kedrogen](#) — CLI tool for scaffolding reproducible data science projects from cookiecutter templates.

MISCELLANEOUS

- [randomizer](#) — Dockerized Shiny application for random allocation of experimental units into treatment groups with specified proportions, featuring SRM testing and covariate balance diagnostics via Love plots.
- Developed multiple [Quarto](#) extensions to enhance scientific documents and presentations, including tools for code highlighting ([line-highlight](#)), interactive SQL demonstrations ([interactive-sql](#)), embedding downloadable resources ([downloadthis](#)), and presentation styling ([reveal-header](#)), available via the [Quarto Extension Registry](#).
- [python-uv-gcloud](#) — Minimal Docker base image with Python, [uv](#), and Google Cloud SDK for CI/CD pipelines and cloud workflows.
- [uvshot](#) — Bash scripts for reproducible and isolated Python environment setup using [uv](#).

TECHNICAL SKILLS

- **Programming:** R, Python, SQL, Lua, Octave, Julia.
- **Statistical Software:** Stata, SAS, Minitab, SPSS.
- **Machine Learning:** Ensemble Models, Causal ML (DML, GRF, BCF), Model Calibration, Conformal Prediction.
- **Data & MLOps:** Shiny, BigQuery, Google Data Studio, GCP, MixPanel, dbt, Kedro, MLflow, Docker, Bash.
- **Others:** $\text{\LaTeX}$, Quarto, Rmarkdown, Git, GitHub Actions, GitLab CI.